Problem Set

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Problem Set Objectives

The weekly problem sets are designed to do the following:

- ...
- ...
- ...

Instructions

Here are some instructions for this problem set. Please read them carefully!

...

Question 1

The `artwork_sample.csv` file contains data for a sample of pieces of art owned by the Tate Art Museum. While more variables are available on the Tate Art Museum's site (github.com/tategallery/collection), you will only be working with the variables featured in the `artwork_sample.csv` file (see below). Save this data object in an R object called `artwork`.

- `id`: Unique ID for each piece of artwork
- `artist`: Name of the artist
- `title`: Title of the artwork
- `type`: Medium used
- `year`: Year the artwork was created
- `width`: width of the artwork, in mm
- `height`: height of the artwork, in mm
- `units`: measurement units for width and height of the artwork
- `area`: surface area (in squared cm)

```
artwork <- read_csv("artwork_sample.csv")
```

(a) Use the `glimpse()` function to view properties of the `artwork` data set. How many observations does it include? How many variables are measured for each observation? Replace `NULL` with your answers in the code chunk below to save your answers in `Q1a_number_of_variables` and `Q1a_number_of_observations`.

```
glimpse(artwork)
```

```
Rows: 1,000
Columns: 10
$ id              <dbl> 52628, 8756, 63026, 43485, 52118, 52634, 52755, 54884,~
$ artist          <chr> "Turner, Joseph Mallord William", "LeWitt, Sol", "Turn~
$ title           <chr> "The Bridge", "A Square Divided Horizontally and Verti~
$ type            <chr> "Watercolour", "Watercolour", "Watercolour", "Watercol~
$ year            <dbl> 1820, 1982, 1830, 1819, 1830, 1831, 1820, 1835, 1969, ~
$ acquisitionYear <dbl> 1856, 1984, 1856, 1856, 1856, 1856, 1856, 1856, 2009, ~
$ width           <dbl> 300, 607, 242, 259, 140, 307, 152, 181, 364, 163, 219,~
$ height          <dbl> 485, 607, 303, 406, 192, 489, 243, 229, 376, 243, 152,~
$ units           <chr> "mm", "mm", "mm", "mm", "mm", "mm", "mm", "mm", "mm", ~
$ area            <dbl> 1455.00, 3684.49, 733.26, 1051.54, 268.80, 1501.23, 36~
```

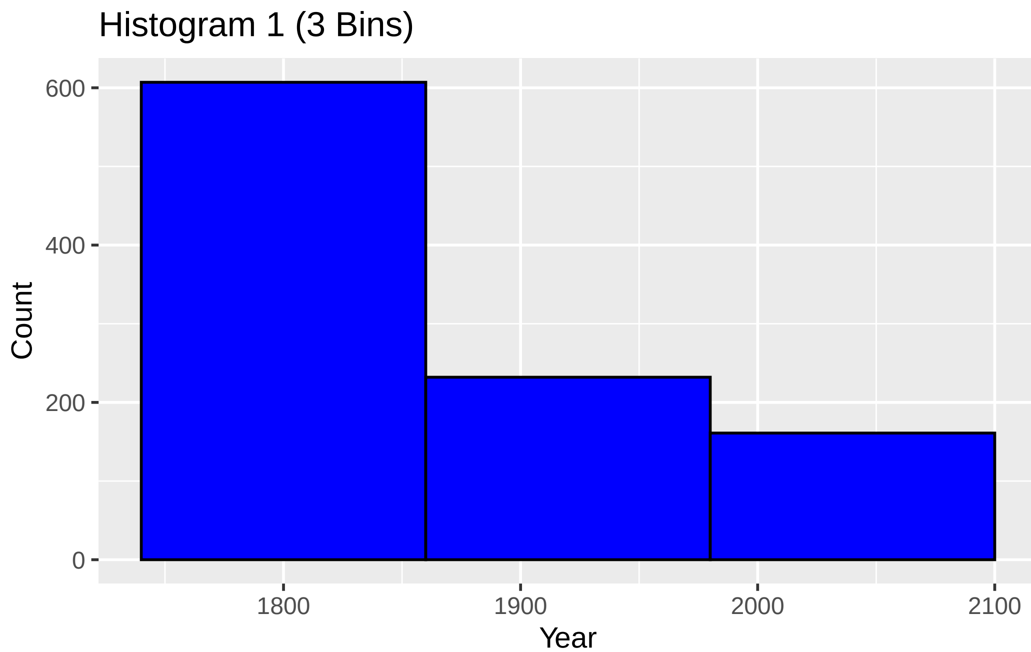
```r
Q1a_number_of_variables <- 10
Q1a_number_of_observations <- 1000
```

Write 1-2 sentences describing the sample using this information and the context.

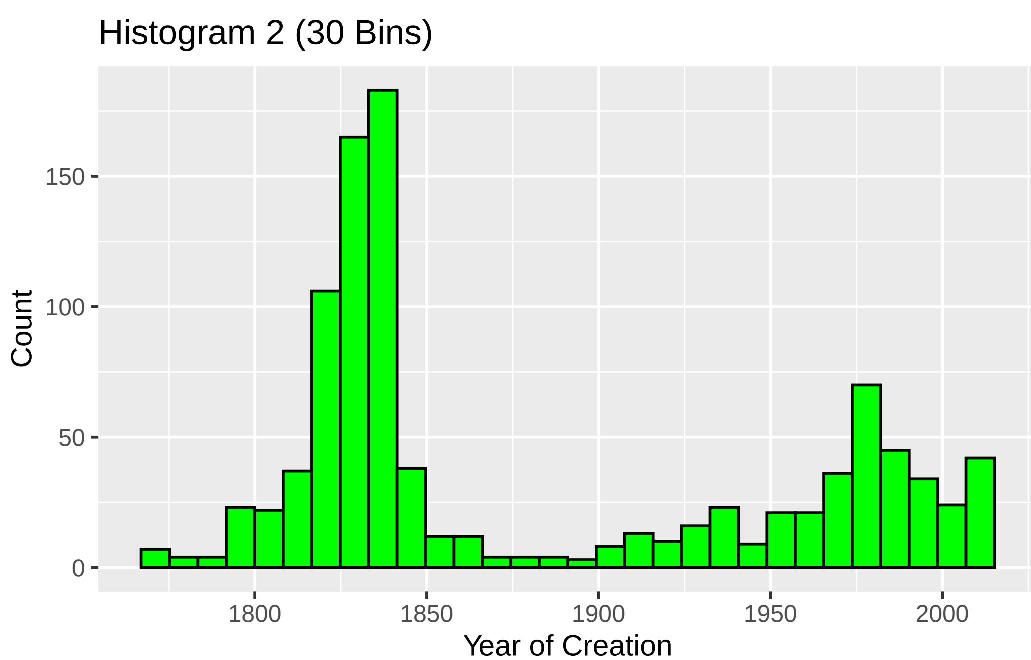
The sample has 10 observations and 1000 variables. Each observation represents a unique artwork with information such as a unique id, title, the artist who created it, and the year the artwork was created.

(b) Create 3 histograms to explore the distribution of years of creation for this sample of pieces of art: (i) one with 3 bins, (ii) one with 30 bins, and (iii) one with 150 bins; make sure to specify meaningful axis labels where appropriate. Which of these histograms is most appropriate to describe the distribution of the artworks' years of creation? Why? Write a few sentences describing the distribution based on the histogram you chose as most appropriate.

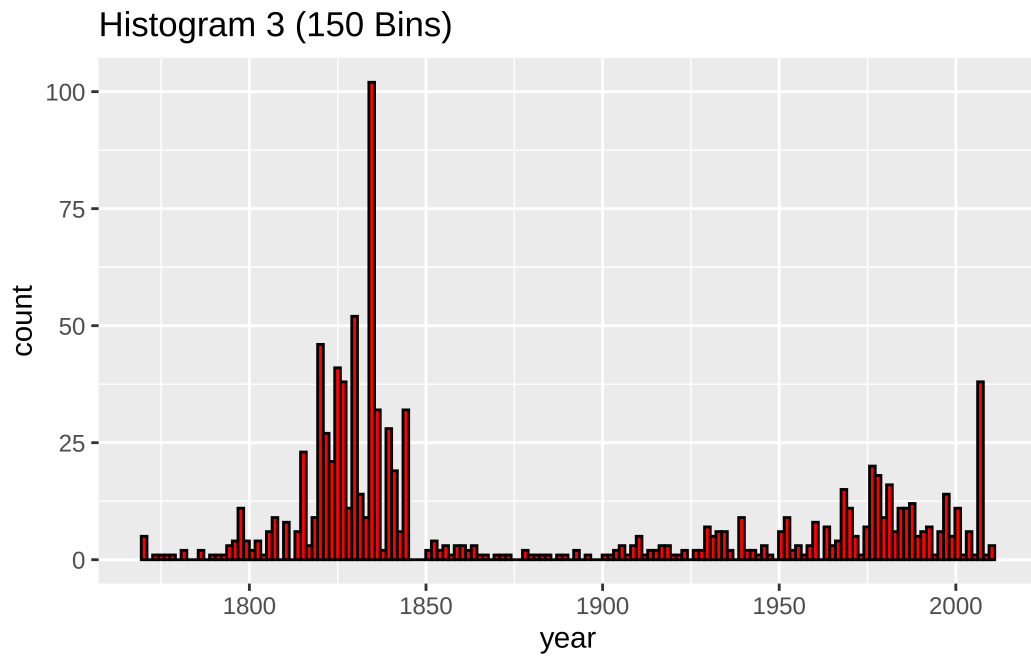
```r
# Create histograms below
Q1b_hist_years_3bins <- ggplot(artwork, aes(x = year)) +
  geom_histogram(bins = 3, fill = "blue", color = "black") +
  labs(title = "Histogram 1 (3 Bins)", x = "Year", y = "Count")

Q1b_hist_years_30bins <- ggplot(artwork, aes(x = year)) +
  geom_histogram(bins = 30, fill = "green", color = "black") +
  labs(title = "Histogram 2 (30 Bins)", x = "Year of Creation", y = "Count")

Q1b_hist_years_150bins <- ggplot(artwork, aes(x = year)) +
  geom_histogram(bins = 150, fill = "red", color = "black") +
  labs(title = "Histogram 3 (150 Bins)", x = "year", y = "count")

print(Q1b_hist_years_3bins)
```




```
print(Q1b_hist_years_30bins)
```




```
print(Q1b_hist_years_150bins)
```



Which histogram is most appropriate to visualize these data?

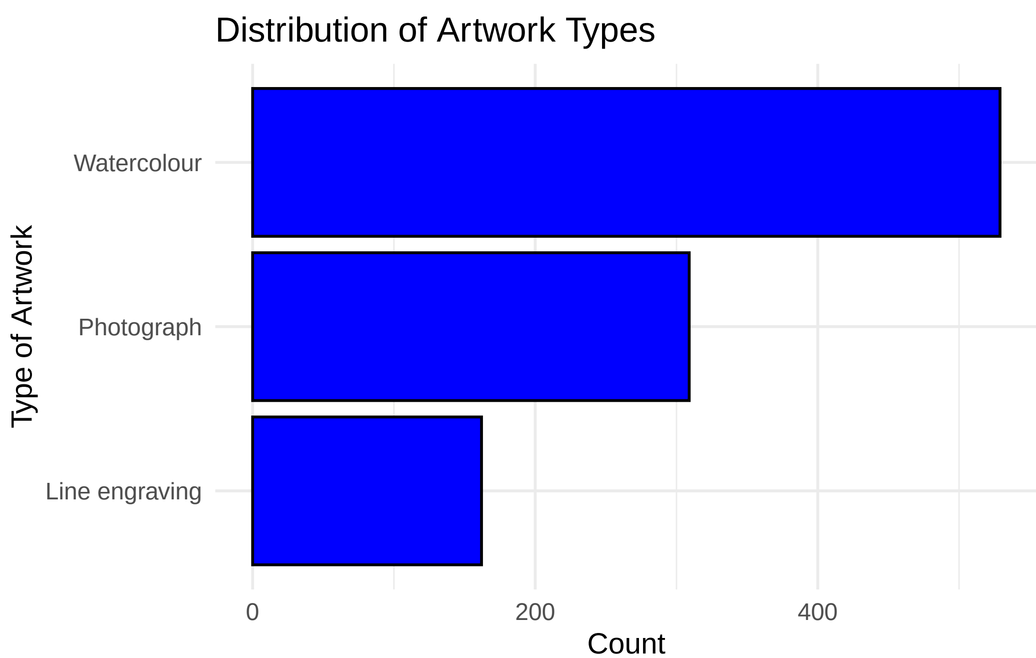
The middle histogram, with 50 bins.

Justify your answer above in a few sentences. Hint: Don't forget to refer to the 3 aspects of quantitative distributions and comment on how each plot lets you visualize each aspect.

The middle histogram, with 50 bins, clearly displays the shape of the distribution, and its centre and spread. The plot lets me visualize each aspect.

(c) Construct a plot to visualize the distribution of the `type` variable. Make sure to specify meaningful axis labels where appropriate. Hint: If you choose a categorical variable with many different categories, you may find it useful to use `coord_flip()` to flip the bars horizontally and/or change the options in the R code chunk to make the plot large (ex: `{r, fig.height=15, fig.width=5}`). From the choices below, select the best description for the distribution of the `type` variable.

```r
# Replace NULL by the code to create your plot and save it in the Q1c_plot
# R object
Q1c_plot <- ggplot(artwork, aes(x = type)) +
  geom_bar(fill = "blue", color = "black") +
  coord_flip() +
  labs(title = "Distribution of Artwork Types",
       x = "Type of Artwork",
       y = "Count") +
  theme_minimal()

Q1c_plot
```



```r
# Among the four descriptions below, decide which one is most accurate
# and replace NULL with "A", "B", "C" or "D" accordingly
Q1c_description <- "C"
```

A: There are about twice as many line engravings than photographs, and almost twice as many photographs than watercolours. The most common type of artwork in this sample of 1000 is watercolour.

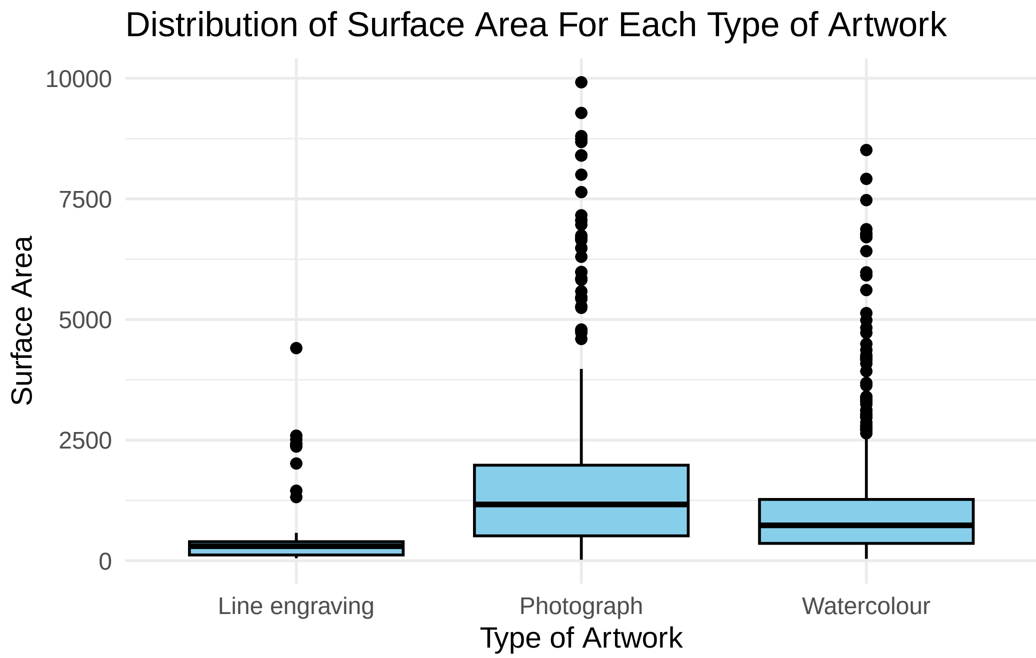
B: There are about twice as many photographs than line engravings, and almost twice as many watercolours than photographs. The most common type of artwork in this sample of 1000 is watercolour.

C: The distribution of artwork type is not symmetrical. The center of the distribution is photograph because it appears in the middle.

D: The distribution of artwork type is left skewed because there are fewer line engravings than photographs and watercolours. The most common type of artwork in this sample is waterclour.

(d) Construct a set of three boxplots showing visual summaries of the distribution of surface area (`area`) for each type of artwork (`type`); make sure to specify meaningful axis labels where appropriate. Save your figure in an R object called `Q1d_plots`. The 3 boxplots should be side by side, not one on top of the other.

```r
# Create your plots below, in a single R object (that is, a single call to ggplot())
Q1d_plots <- ggplot(artwork, aes(x = type, y = area)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Distribution of Surface Area For Each Type of Artwork",
       x = "Type of Artwork",
       y = "Surface Area") +
  theme_minimal()

Q1d_plots
```

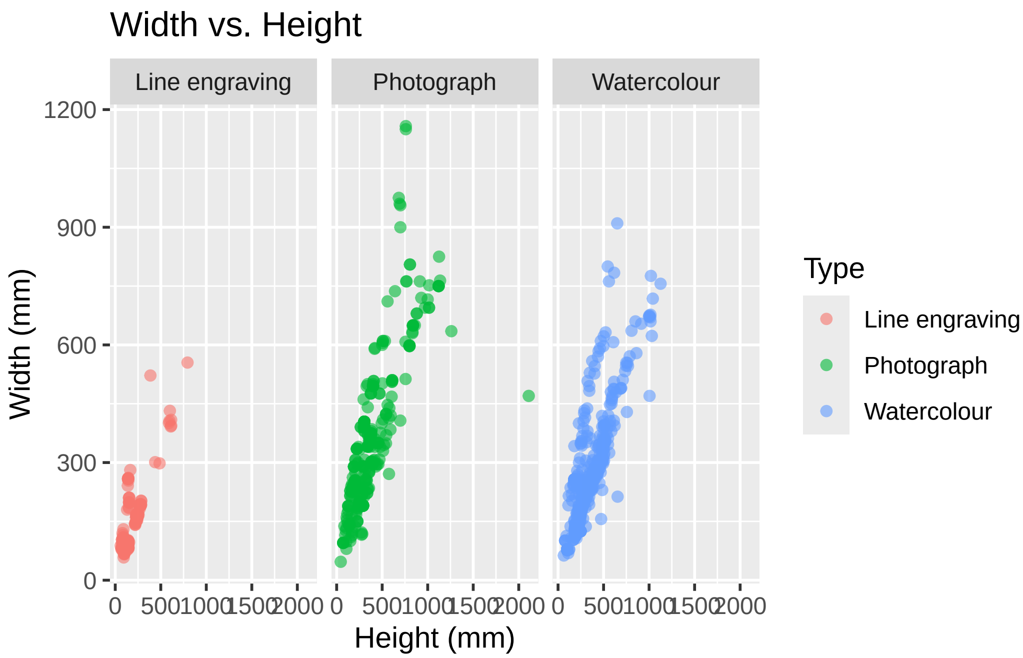


Write 3-4 sentences comparing these distributions.

Watercolours have the smallest range of surface area values and line engravings have the biggest range. Photographs have a larger median. There are more outliers for line engravings and photographs than the other types of artworks.

(e) Make a scatterplot showing the `width` versus `height` for each type of artwork (`type`) using the `facet_wrap()` option; make sure to specify meaningful axis labels where appropriate.

```r
# Create your plot below
Q1e_plots <- ggplot(artwork, aes(x = height, y = width, color = type)) +
  geom_point(alpha = 0.6) +
  facet_wrap(~type) +
  labs(title = "Width vs. Height",
       x = "Height (mm)",
       y = "Width (mm)",
       color = "Type")

Q1e_plots
```



Write 1-3 sentences describing any trends you see.

There seems to be some kind of positive correlation between width and height, meaning artworks that are wider also tend to be taller (at least excluding outliers). There seem to be narrow ranges for watercolours, meaning they are more square.

Conclusion

We hope this exercise helped to emphasize the importance of the ethical considerations we discussed in class when it comes to making, interpreting, and communicating data visualizations.